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From Green Bean to Cup Score

Predicting the cupping score of green coffee with machine learning

Project 2 · Machine Learning: From Data to Predictions
Ironhack · Data Science & Machine Learning

Rizki · October 2026

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What makes a green coffee score higher?

Where the idea started

- A **green bean identifier**: upload a photo, or point a live camera at a bean, and see the information about that specific green bean on screen, live.
- Going further: connect it to **smart glasses** such as the Ray-Ban Meta, to capture a bean and view the coffee information in a virtual or augmented-reality style experience.

What it needs

- A **database** of the characteristics that make up the average green bean, and the average of each specific content of the bean.
- That led us to use the **cup score (Total Cup Points) as the target**, so machine-learning models can learn how a bean's characteristics relate to its quality.

The question for the ML project

Can we predict a green coffee's Total Cup Points from facts known before it is cupped: origin, variety, processing method, altitude, moisture, defects and colour?

Who uses it and for what decision: a buyer deciding which lots to taste first.

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The data

1310 Arabica lots, 20 columns. One row is one sample of green coffee that a grader cupped. The target is its Total.Cup.Points. Source and licence are on the References page.

Raw datasets | EDA & Cleaning | Score distribution | Score by country | Altitude and moisture | Missing values

The original files, before any cleaning. Click a title to open it. The tables scroll, and the column titles stay on top while you scroll down. The datasets are listed alphabetically.

- Coffee Quality Institute (CQI), via GitHub · Arabica · 1,311 rows × 44 columns
- Coffee Quality Institute (CQI), via GitHub · Arabica · 1,312 rows × 53 columns
- Coffee Quality Institute (CQI), via GitHub · Robusta · 28 rows × 44 columns
- Coffee Quality Institute (CQI), via GitHub · Robusta · 28 rows × 51 columns
- FAOSTAT · Food and Agriculture Organization (FAO) · 1,146 rows × 15 columns
- USDA Foreign Agricultural Service · PSD Online · 87,704 rows × 12 columns

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Metric and baseline

Metric: MAE, the average miss in cup points. It is easy to explain and is not dominated by a few extreme scores. R^2 (the share of the variation we explain) is reported next to it.

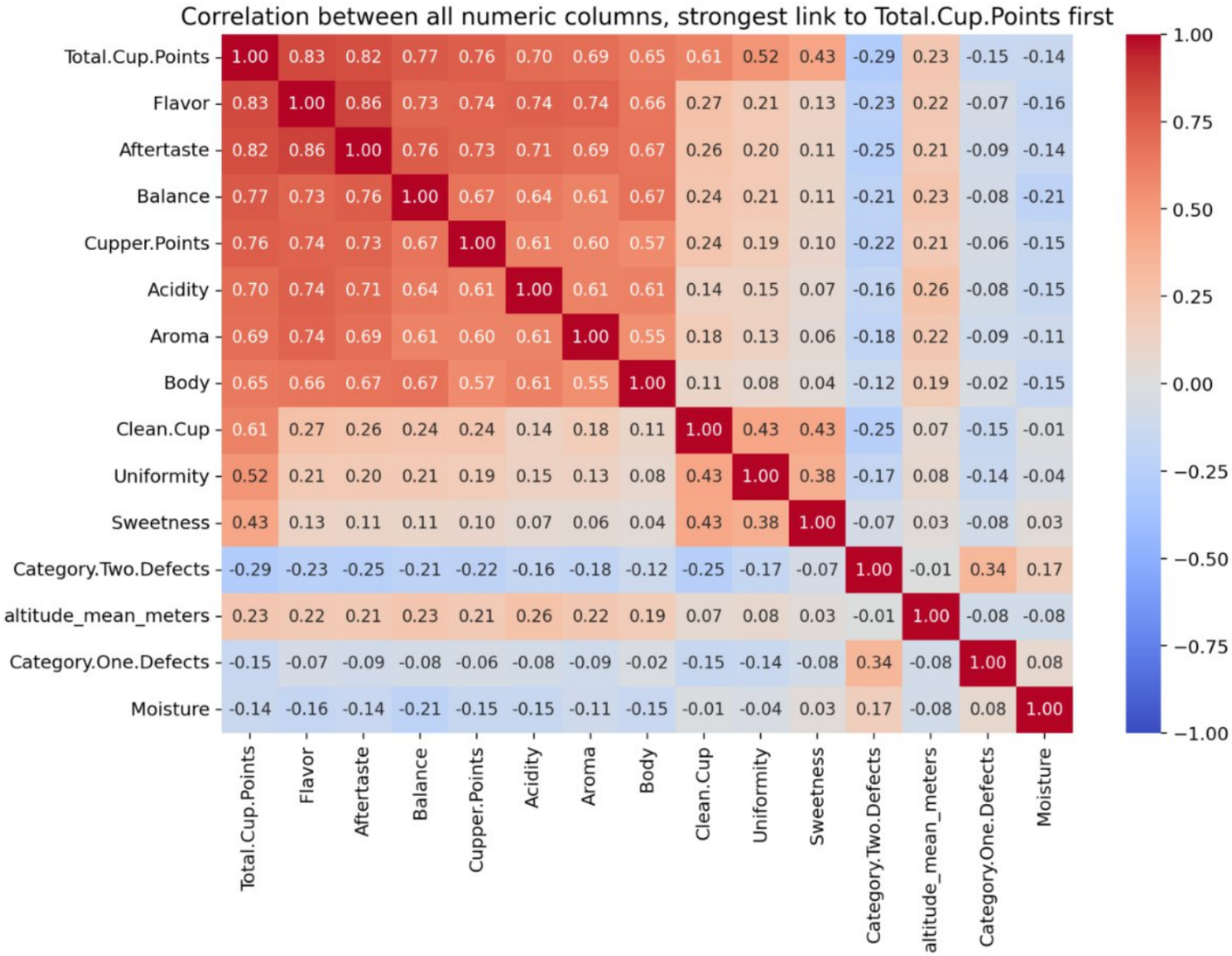
Which mistake costs more: specialty coffee starts at about 80 points and most scores cluster just above it, so a miss of 1 to 2 points can move many coffees across that line.

The baseline always guesses the average cup score. Its MAE and the Huber MAE are on the next slides.

› Show the code for the baseline and the cross-validation

Correlation heatmap | Regression Evaluation | Classification Evaluation

How every numeric column moves together, with the columns ordered by how strongly they move with Total.Cup.Points.



The ten part scores add up to the total, so they correlate strongly with it by construction. The model does not use them, only the bean and origin measurements.

› Show the code behind this diagram

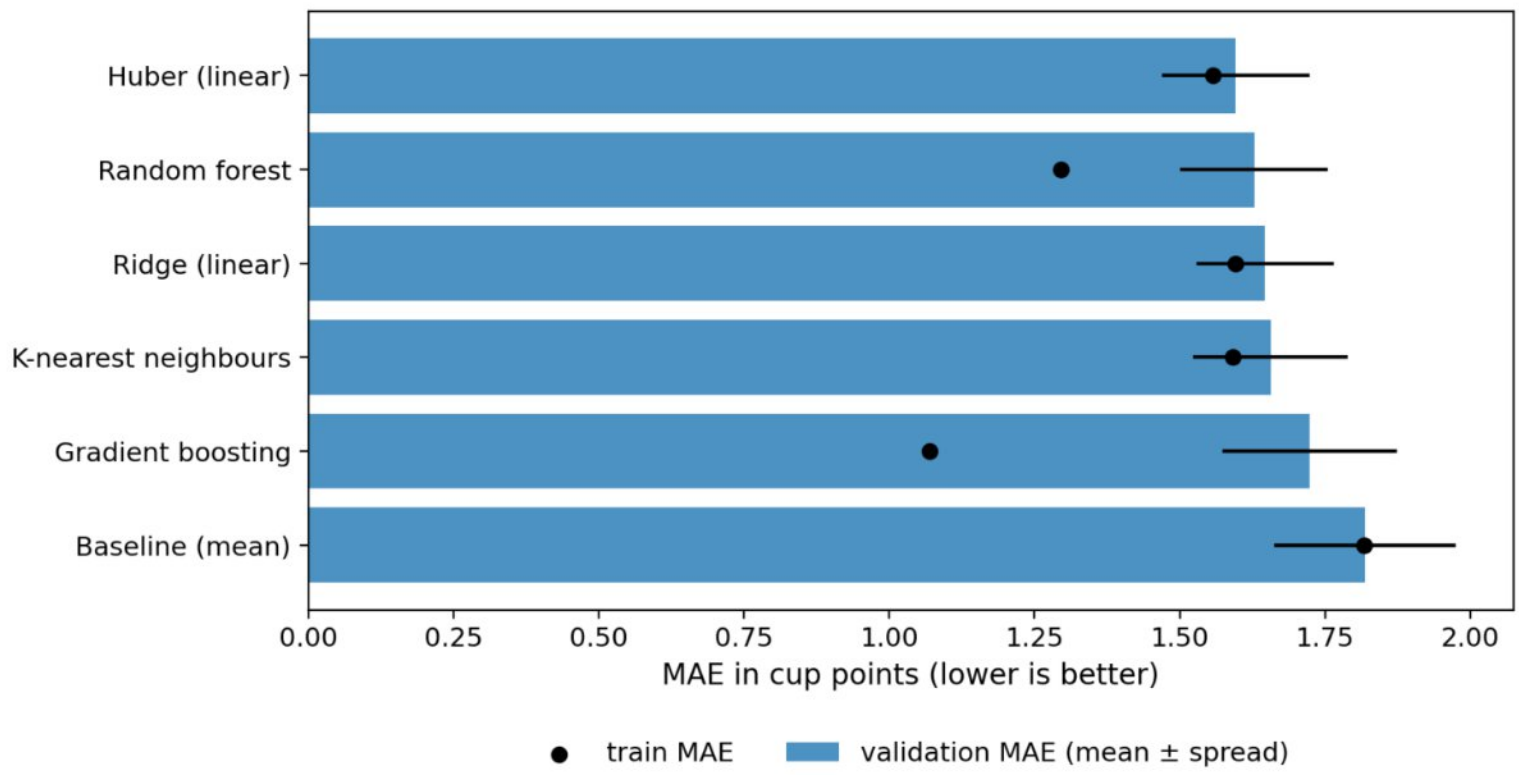
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What I tried

Huber (linear)	Random forest	Ridge (linear)	K-nearest neighbours	Gradient boosting	Baseline (mean)
1.596	1.628	1.647	1.656	1.724	1.819

Validation MAE in cup points (lower is better), the mean over the same 5 cross-validation folds. Tuned on the training lots: random forest 1.597 (best of 48 settings), gradient boosting 1.633.

Same 5 cross-validation folds for every model. The bar is the validation MAE with its spread, the black dot is the training MAE.

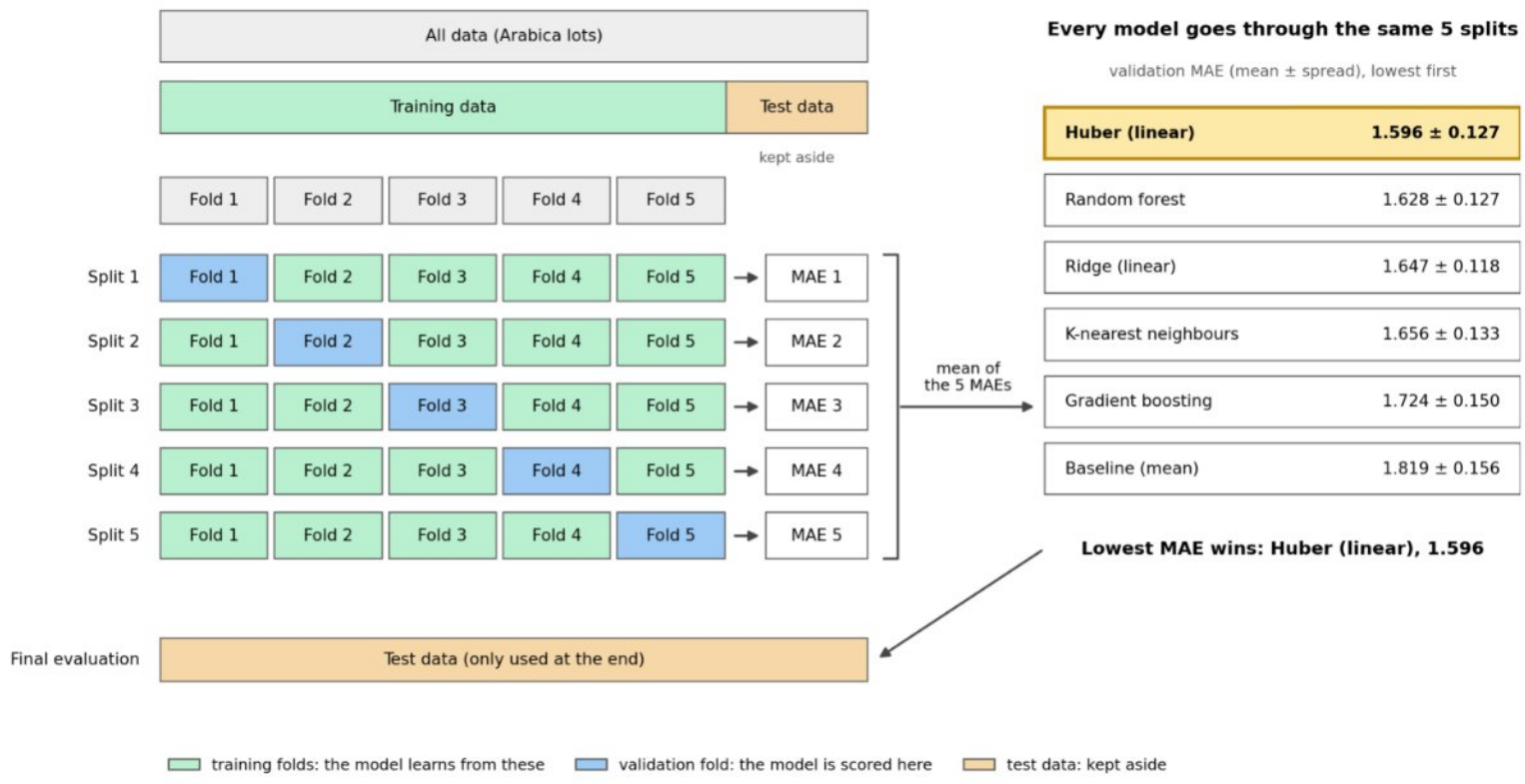


The biggest gap between train and validation is Gradient boosting: the model is memorising the training lots.

› Show the code behind this diagram

How each model is tested

Every model is tested the same way, so the comparison is fair. The test lots are put aside first. The training lots are cut into 5 folds. In each of the 5 splits one fold (blue) is held back and the model learns from the other four (green), then it is scored on the held-back fold. That gives 5 MAE values, and their mean is the validation MAE of the model. The model with the lowest validation MAE is the one we choose. The test lots are only used at the end, for the final evaluation.

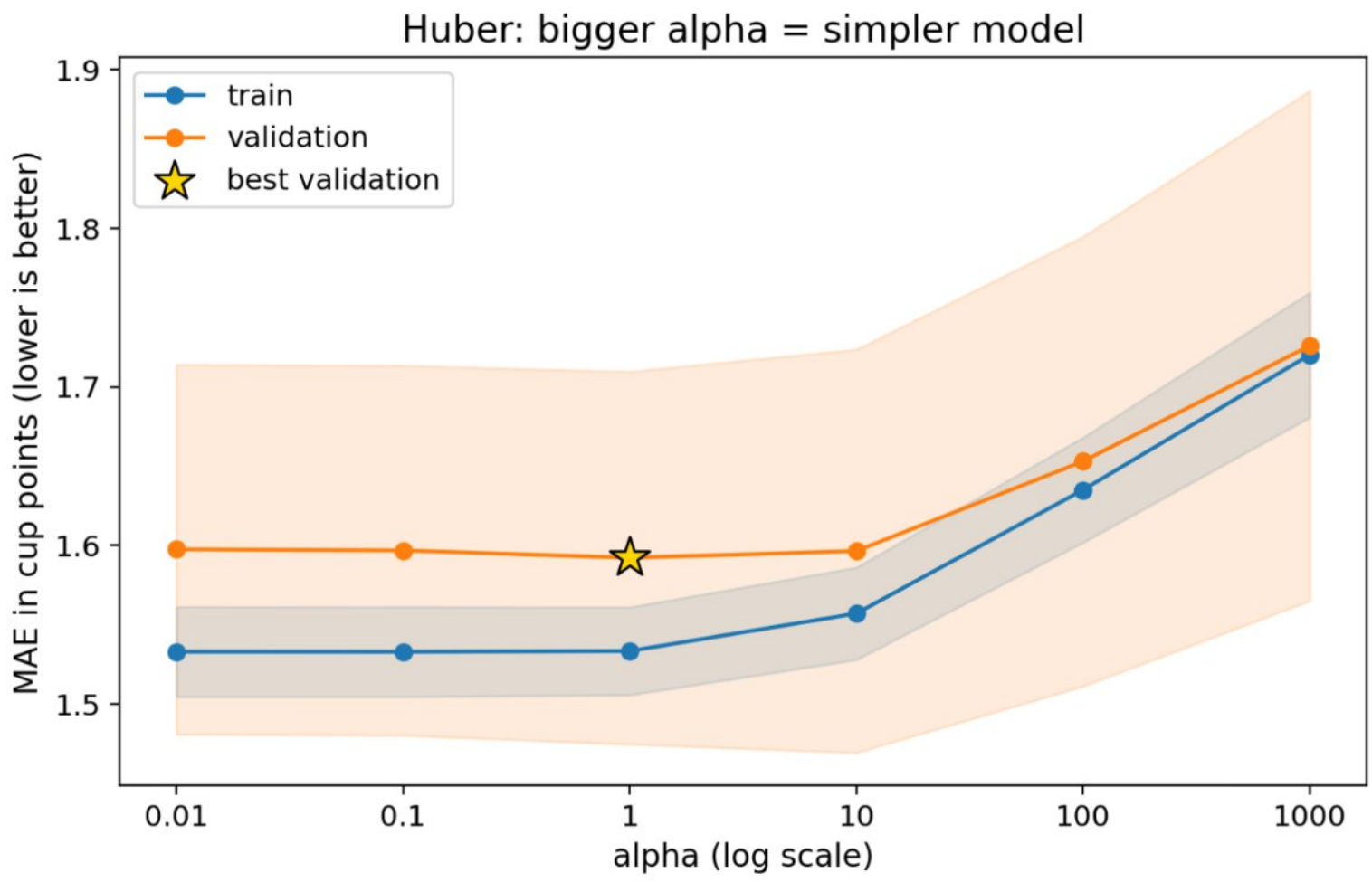


The same 5 splits are used for every model, so a difference in MAE comes from the model and not from the luck of the split. The numbers on the right are the validation MAE and its spread from the bar chart above.

› Show the code behind this diagram

The final model: Huber regression, alpha = 10

Huber is a linear model that counts small misses by their square but big misses only by their size, so a few very low scores pull the line around less. In cross-validation it beat Ridge on all 5 folds (MAE 1.596 against 1.647) and tied the tuned random forest (1.597, the best of 48 settings tried), so I chose the linear model I can explain. On the test lots the two had almost the same MAE.



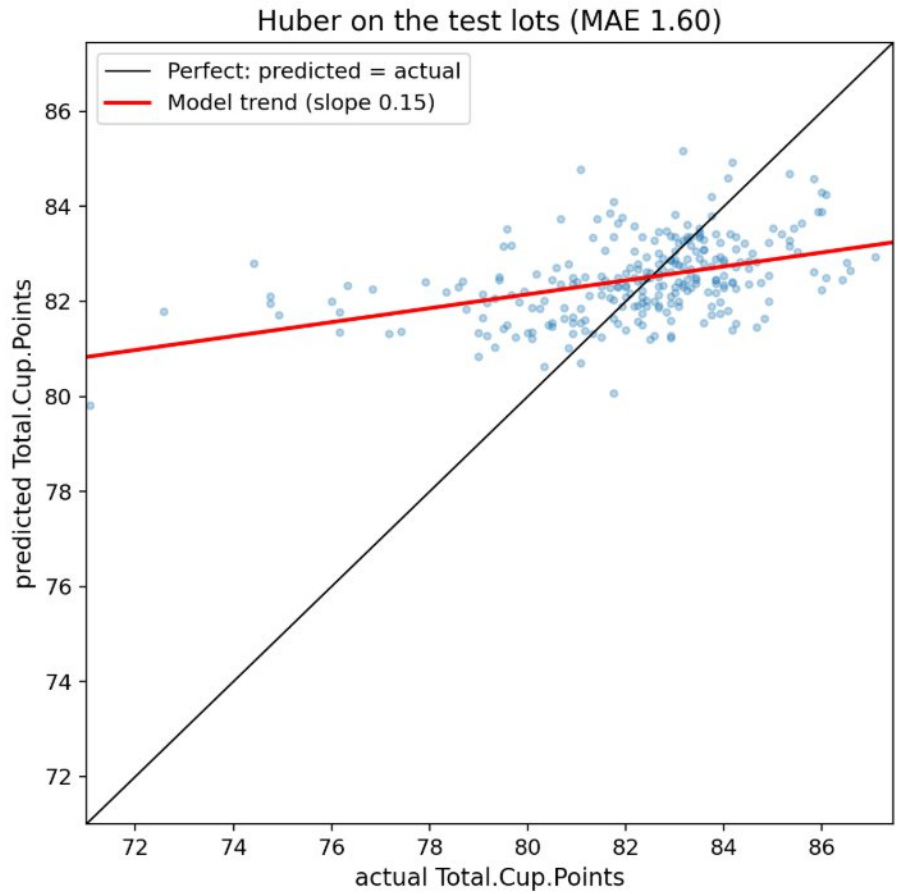
› Show the code behind this diagram

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The final model

Baseline MAE (cross-validated)	Huber MAE (cross-validated)	Huber MAE on the test lots
1.82	1.60	1.60

R^2 on the test lots: 0.17. 262 test lots, split with random_state=42. These numbers should match notebook 03. The test lots were scored for both Ridge and Huber; Huber was chosen from the cross-validation.

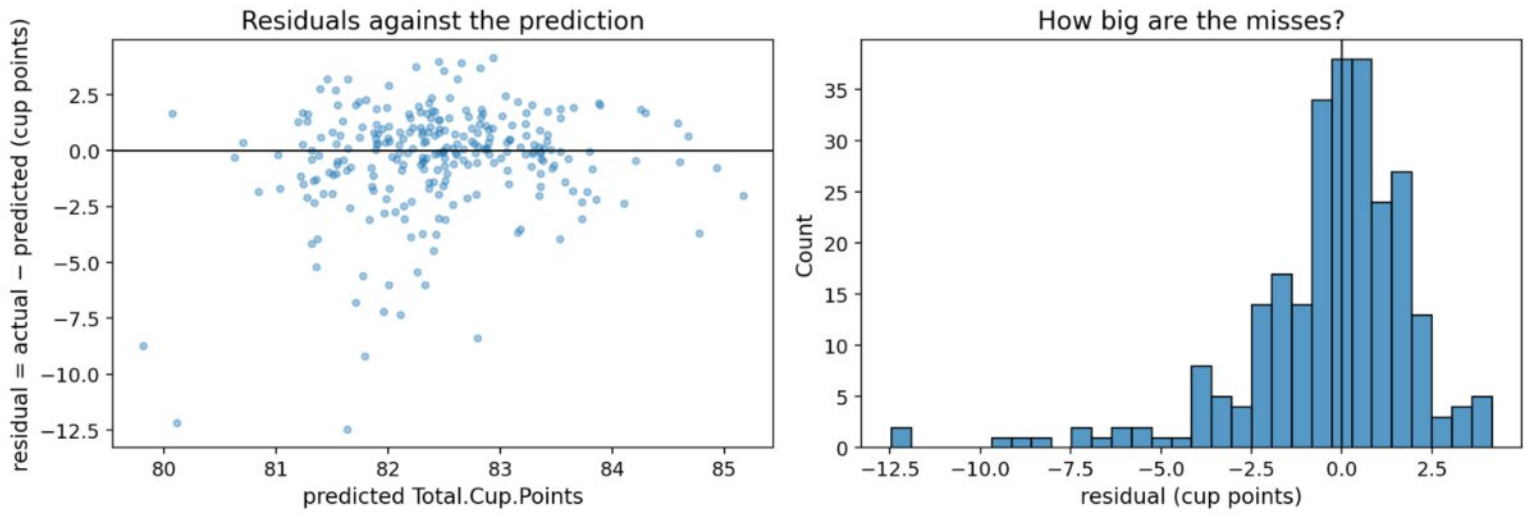


The closer the model's trend is to the perfect line, the better the predictions. A flatter trend means the model pulls every lot toward the average score.

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Where it goes wrong



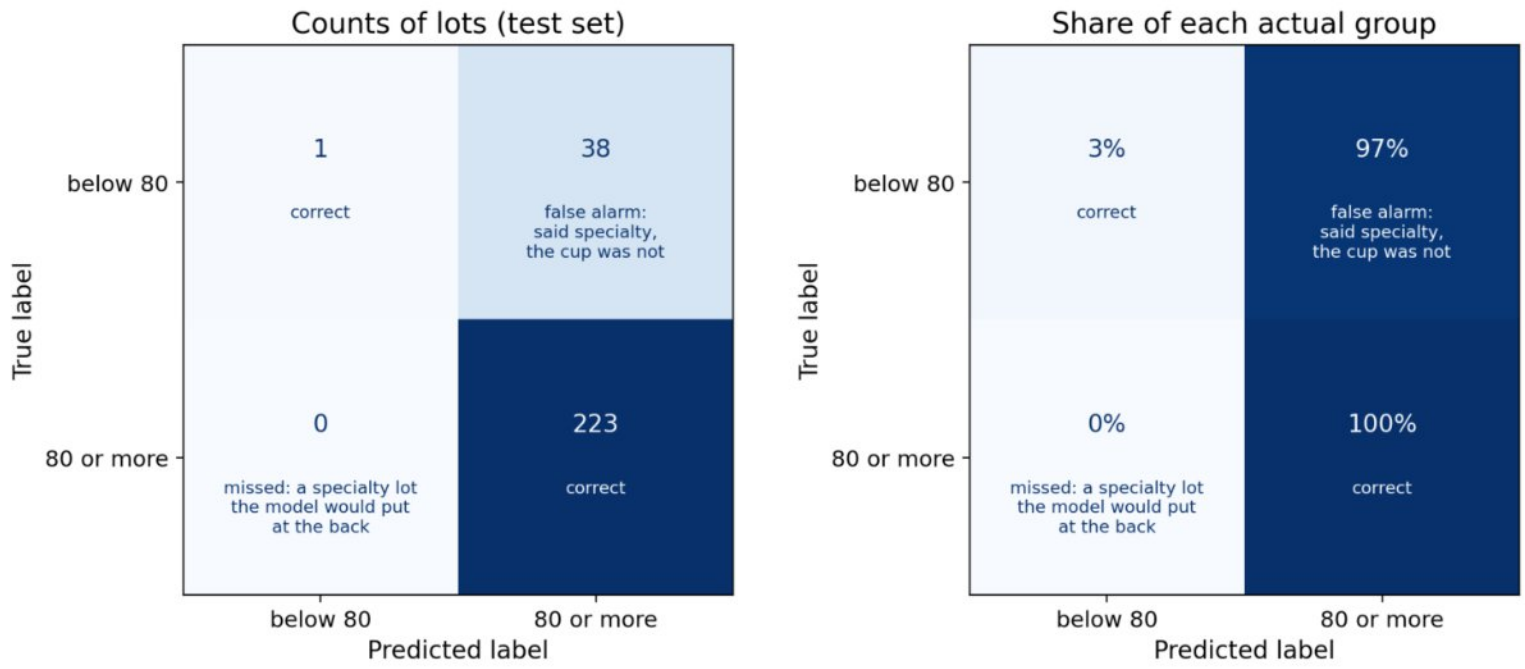
› Show the code behind this diagram

The typical miss is 1.60 cup points, and the biggest miss on the test lots is 12.5 points.

Business reading: use it to decide what to taste first, not to screen lots out.

The 80-point line

Right side of the 80 line: **85.5%**. Always saying 'specialty' with no model at all: 85.1%. Sub-80 lots caught: 3%. Flagged lots that really are sub-80: 100%.



› Show the code behind this diagram

Error by country

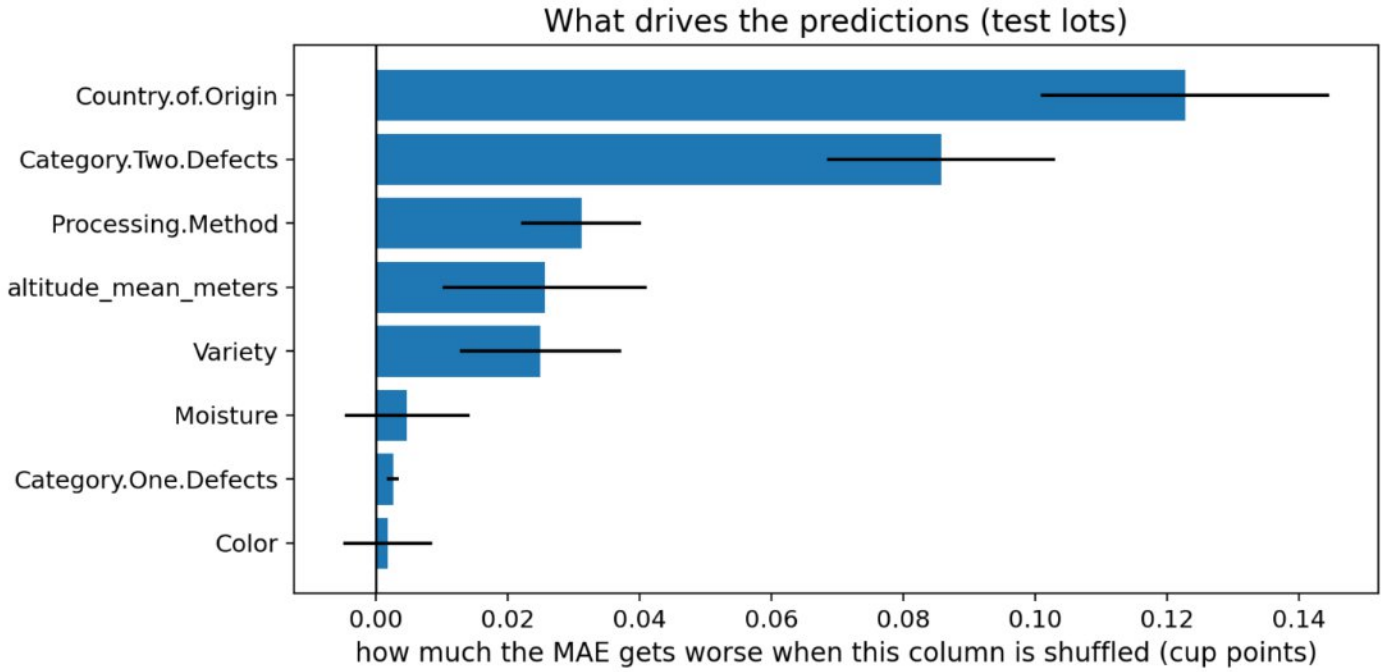
group	lots	MAE	bias
Honduras	12	2.27	-1.7
Guatemala	35	2.07	-1.1
United States (Hawaii)	20	1.98	-1.01
Mexico	39	1.71	-0.57
Tanzania, United Repub	10	1.42	0.2
Indonesia	10	1.38	-0.87
Taiwan	22	1.38	0.63
Brazil	20	1.11	0.92
Colombia	45	0.99	-0.26

› Show the code behind this table

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What drives the predictions



Each column is shuffled in turn to see how much the average miss grows. Read with care: this shows what the model uses, not what causes a high score.

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GCCS Predictor

Score a new coffee | Check a real test lot

Country of origin: Brazil

Variety: Arusha

Processing method: Natural / Dry

Bean colour: Blue-Green

Altitude is known

Mean altitude (m): 1200

Moisture was measured

Moisture: 0,11

Category one defects: 0

Category two defects: 0

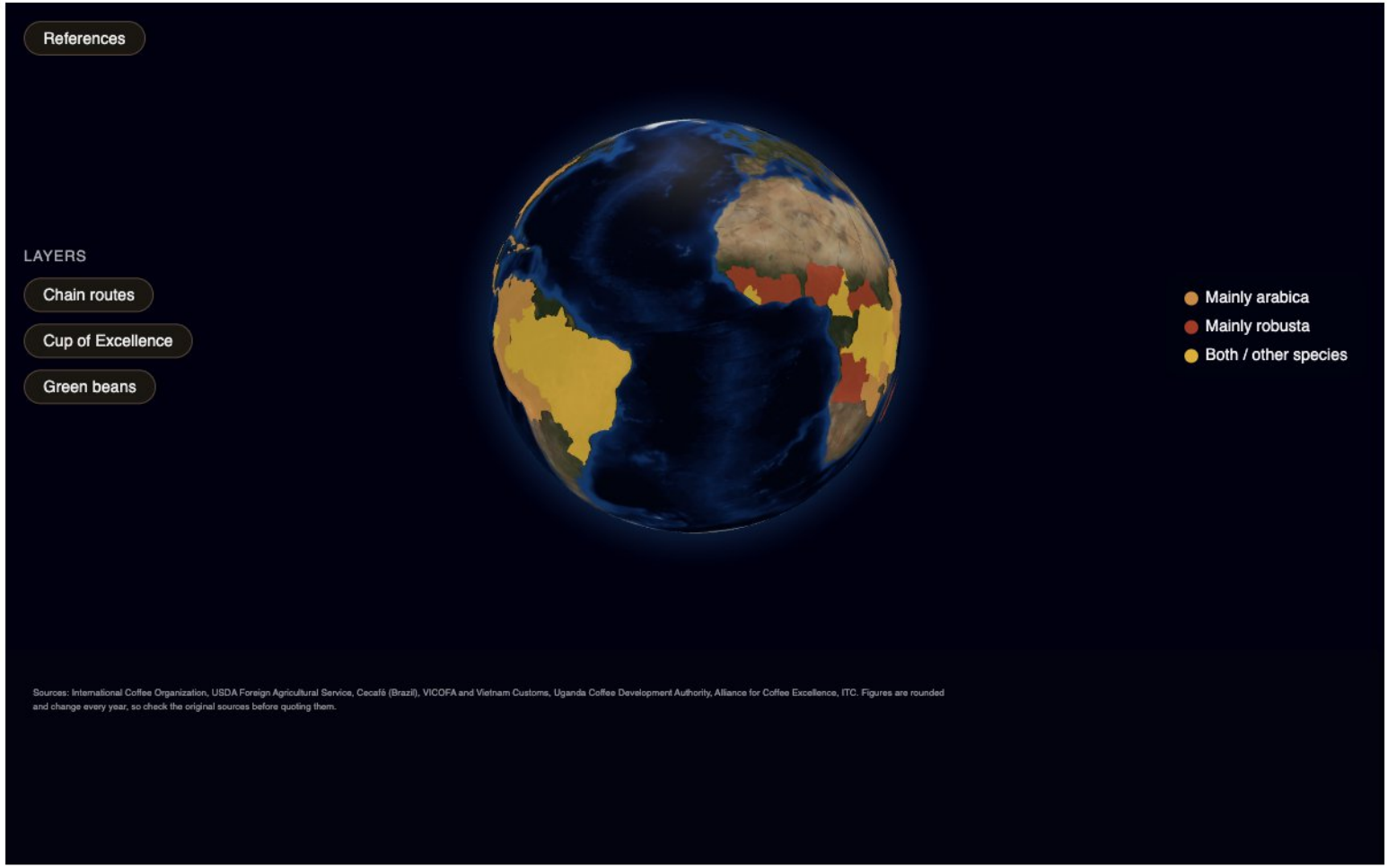
Predict cup score

› Show the code behind this prediction

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Coffee globe



The globe loads its map from the internet, so check your wifi before presenting.

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Limits and conclusions

The question, answered

Can we predict a green coffee's Total Cup Points from facts known before it is cupped? A careful yes. On the test lots the typical miss is 1.60 cup points, against 1.83 for always guessing the average, so the model is 0.23 points better (12%). R^2 is 0.17: origin, variety, processing, altitude, moisture, defects and colour explain about 17% of the differences between lots. The rest is in the cup.

What makes a green coffee score higher?

The model leans most on **Country of Origin**, **Category Two Defects** and **Processing Method**, the columns whose shuffling hurts the predictions most. This shows what the model uses, not what causes a high score. The ten cupping part scores are not used: they are the taster's own answers and add up to the total.

Limits

- **Imbalanced data.** 1,310 Arabica lots from the CQI reviews, below the brief's suggested few thousand rows. Coverage is uneven: Mexico has 236 lots, Indonesia 20.
- **Gaps.** 23% of lots have no usable moisture and 20% no altitude (a moisture of exactly 0.0 was most likely a missing value in disguise). The pipeline fills them from the training lots.

Would I use it?

Yes, to decide what to taste first, not to screen lots out. A typical miss of 1.60 points is enough to move a lot across the 80 line, so the model should rank lots, not pass or fail them.

What's next

The idea that started this project is a **green bean identifier**: point a camera at a bean and see what is known about it. This project shows that a cup score can be predicted from the facts about a lot, but a camera needs something this data does not have: photos.

To move closer to that goal I would need to go through labelling and recording thousands of green beans and save them to the database: a photo of each bean, its origin, variety, processing, altitude, moisture, defects and colour, and its cup score. The CQI file has the facts and the scores, but no photos.

1. **Build the database.** Record and label thousands of green beans, with moisture and altitude measured and not left empty. This answers the two biggest limits above: too few lots and too many gaps.
2. **Cover more origins.** Add lots from origins the data barely has (Indonesia has 20) and more Robusta, so the model is not Arabica-only.
3. **Teach the camera what it can see.** Train an image model on the photos to read colour, visible defects and size. Origin, altitude and moisture cannot be seen, so they come from the database or the supplier.
4. **Rerun the models on the bigger data.** Try the random forest and gradient boosting again, keep Huber only if it still wins, and check whether the error keeps falling.
5. **Build the identifier.** Upload a photo or point a live camera at a bean and see its information on screen, then try smart glasses such as the Ray-Ban Meta.

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Questions?

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References

56 sources, the same list as the References button on the globe: dataset 11, paper 32, image set 1, tool 6, web page 2, trade press 4.

› Dataset (11)

› Paper (32)

› Image set (1)

› Tool (6)

› Web page (2)

› Trade press (4)